

10. A fixed mass of monatomic ideal gas at a volume V at temperature T and pressure P has a root mean square speed c . After absorbing heat, its final volume is V and final pressure is $2P$.

What is the new root mean square speed of the ideal gas?

A $0.5 c$

B $0.71 c$

C $1.4 c$

D $2.0 c$

Ans: C

$pV = NkT$. Since volume V remains the same and pressure is doubled, it means that the final temperature is also doubled.

$$\text{Initial: } \frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} k T$$

$$\text{Final: } \frac{1}{2} m \langle c_f^2 \rangle = \frac{3}{2} k (2T)$$

$$\begin{aligned} \sqrt{\langle c_f^2 \rangle} &= \sqrt{2} \sqrt{\langle c^2 \rangle} \\ &= 1.4 \sqrt{\langle c^2 \rangle} \end{aligned}$$

11 An ideal gas is confined in a cylinder fitted with a frictionless piston. When 90 J of heat